# System Analysis and Design

**Eighth Edition**

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## Appendix 5A

Normalizing the Data Model

# Validating an Entity Relationship Diagram

- Creating ERDs is pretty tough
- There are some general design guidelines that you can keep in mind as you build ERDs
- Once the ERDs are drawn, you can use a technique called ***normalization*** to validate that your models are well formed

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# Design Guidelines

- Best practices rather than rigid rules
- Entities should have many occurrences
- Avoid unnecessary attributes
- Clearly label all components
- Apply correct cardinality and modality
- Break attributes into lowest level needed
- Labels should reflect common business terms
- Assumptions should be clearly stated

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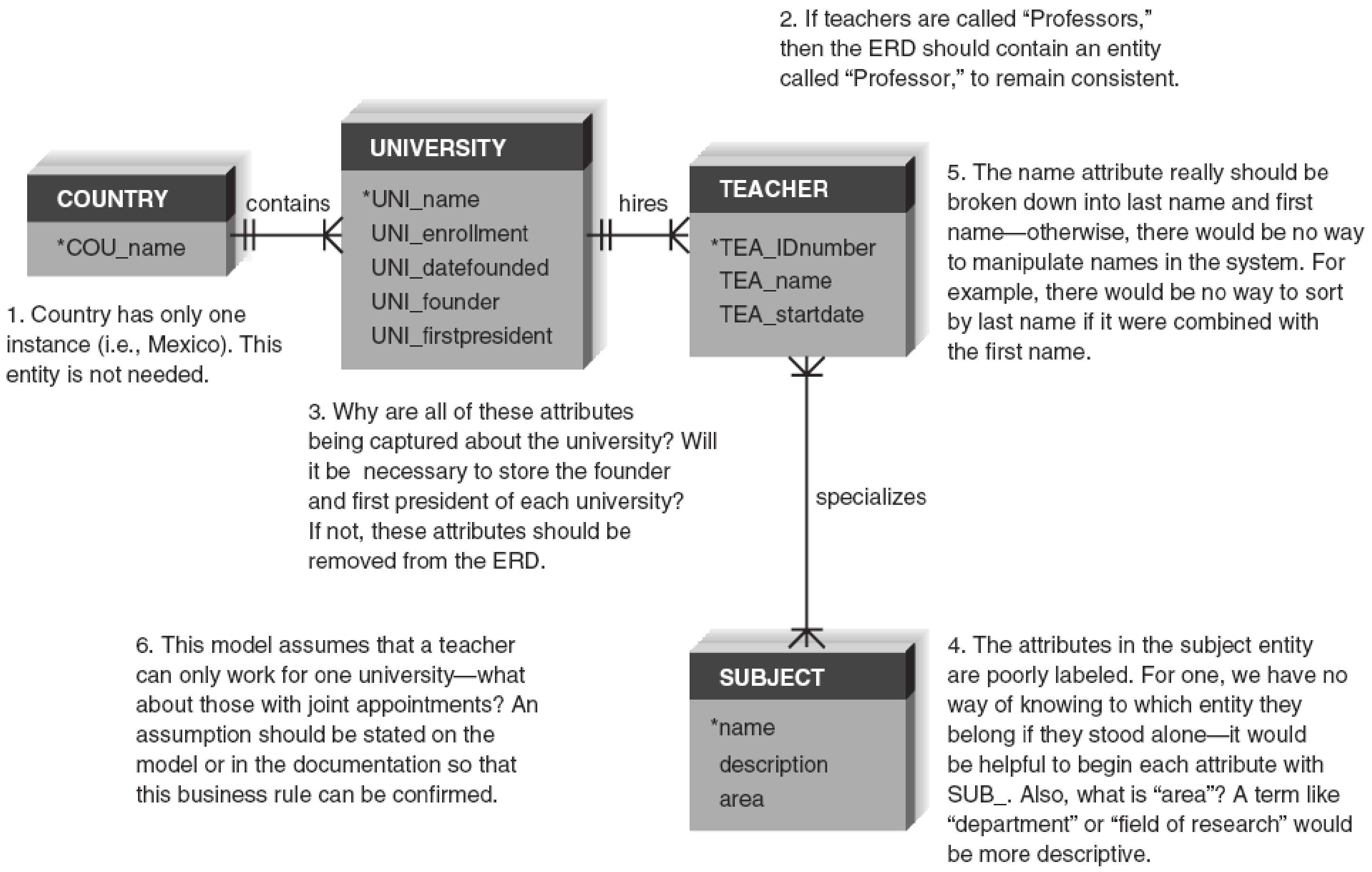
# Data Modeling Guidelines Summary

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# Normalization

- ***Normalization*** is a process whereby a series of rules are applied to a logical data model or a fi le to determine how well formed it is
- Normalization rules help analysts identify entities that are not represented correctly in a logical data model, or entities that can be broken out from a file
- Three normalization rules described in the appendix

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# Balancing ERDs with Data Flow Diagrams

- All analysis activities are interrelated
- Process models contain two data components
- Data flows and data stores
- The D F D data components need to balance the ERD’s data stores (entities) and data elements (attributes)
- Many CASE tools provide features to check for imbalance
- Check that all data stores and elements correspond between models
- Data that is not used is unnecessary
- Data that has been omitted results in an incomplete system
- Do not follow thoughtlessly -- check that the models make sense!

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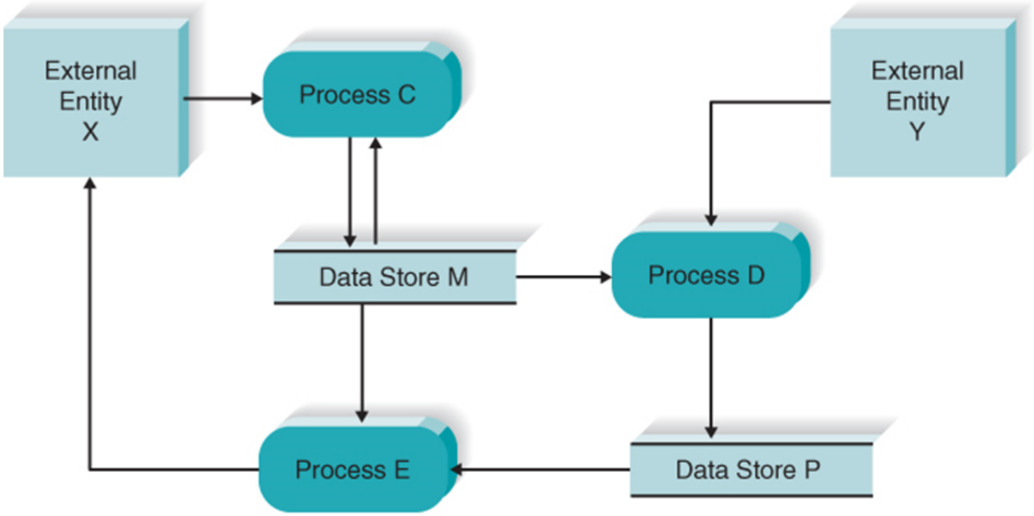
# Use of a CRUD Matrix




| | Process C | Process D | Process E |
|---|---|---|---|
| Data Entity M | | | |
| Attribute M-1 | CRUD | R | R |
| Attribute M-2 | CRUD | | R |
| Attribute M-3 | CRUD | R | |
| Attribute M-4 | CRUD | | R |
| Data Entity P | | | |
| Attribute P-1 | | C | R |
| Attribute P-2 | | C | |
| Attribute P-3 | | C | R |

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# Normalization

- Technique used to validate data models
- Series of rules applied to logical data model to improve its organization
- Three normalization rules are common
  1. First normal form (1NF)
  2. Second normal form (2NF)
  3. Third normal form (3NF)

# Summary of Normalization Steps

| | |
|---|---|
| | 0 Normal Form |
| Do any attributes have multiple values for a single instance of an entity? | Yes: Remove the repeating attributes and repeating groups. Create an entity that describes the attributes. Usually, you will need to add a relationship to connect the old and new entities.<br>No: The data model is in 1NF. |
| | 1 Normal Form |
| Is the identifier composed of more than one attribute? If so, are any attribute values dependent on just part of the identifier? | Yes: Remove the partial dependency. Move the attributes to an entity in which their values are dependent on the entire identifier. Usually, you will need to create a new entity and add a relationship to connect the old and new entities.<br>No: The data model is in 2NF. |
| | 2 Normal Form |
| Do any attribute values depend on an attribute that is not the entity's identifier? | Yes: Remove the transitive dependency or derived attribute. Move the attributes to an entity in which their values are dependent on the identifier. Usually, you will need to create a new entity and add a relationship to connect the old and new entities.<br>No: The data model is in 3NF. |
| | 3 Normal Form |

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# First Normal Form

- A logical data model is in ***first normal form*** (1NF) if it does not contain attributes that have repeating values for a single instance of an entity
- Often, this problem is called ***repeating attributes***, or ***repeating attribute groups***

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# Example: Unnormalized Entity

- Begin with an entity from the logical data model
- Do any attributes (or groups of attributes) occur more than once for a single occurrence of the entity?
- Yes

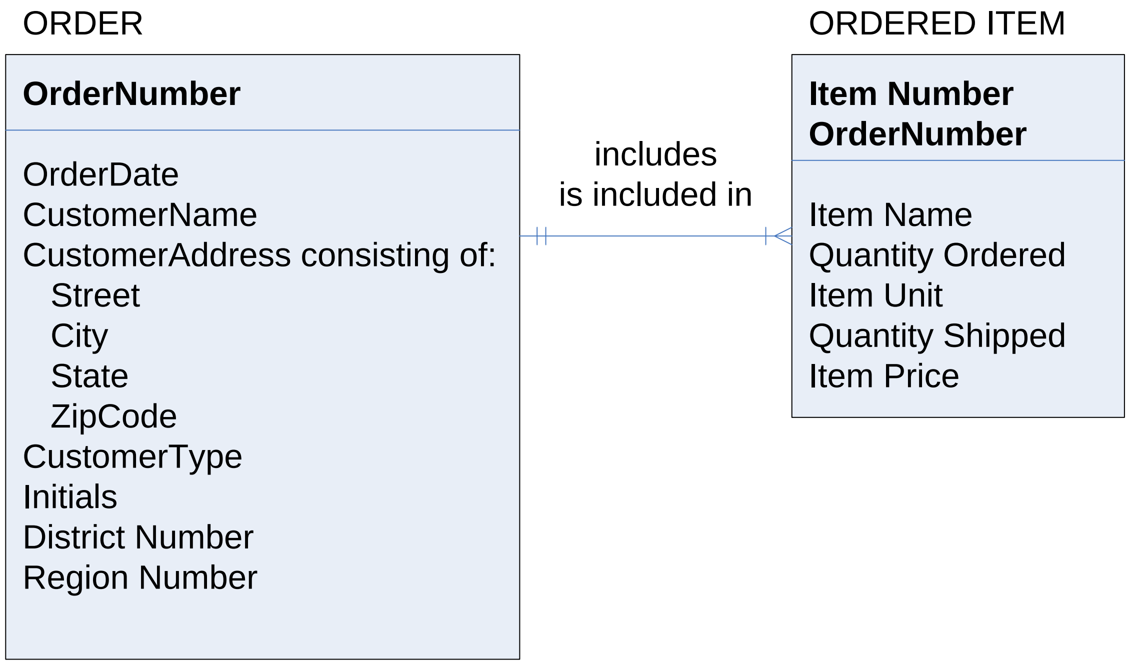
ORDER

| OrderNumber |
|---|
| OrderDate<br>CustomerName<br>CustomerAddress consisting of:<br>  Street<br>  City<br>  State<br>  ZipCode<br>CustomerType<br>Initials<br>District Number<br>Region Number<br>1 to 22 Occurrences of:<br>  Item Number<br>  Item Name<br>  Quantity Ordered<br>  Item Unit<br>  Quantity Shipped<br>  Item Price |

Yes

# Example: First Normal Form

- Do any attributes (or groups of attributes) occur more than once for a single occurrence of the entity?
- If yes, move the attributes (or groups) into separate entities

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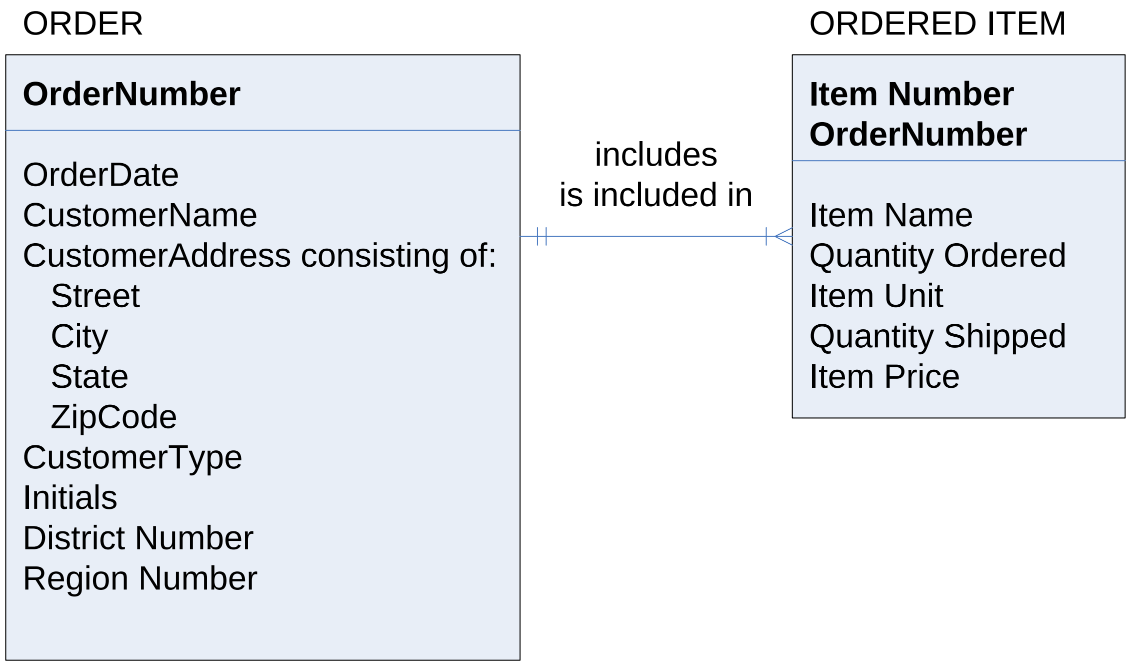
# Potential Anomalies with First Normal Form

- Insert anomaly:
  - Insert a new Item?
  - Can not do without Order Number
- Deletion anomaly:
  - Assume only one order for Item #456
  - What happens if order is cancelled?
  - Will lose all information about Item #456
- Update anomaly:
  - The price of Item # 789 changes
  - What problem occurs?
  - Need to search entire database for all occurrences of Item # 789

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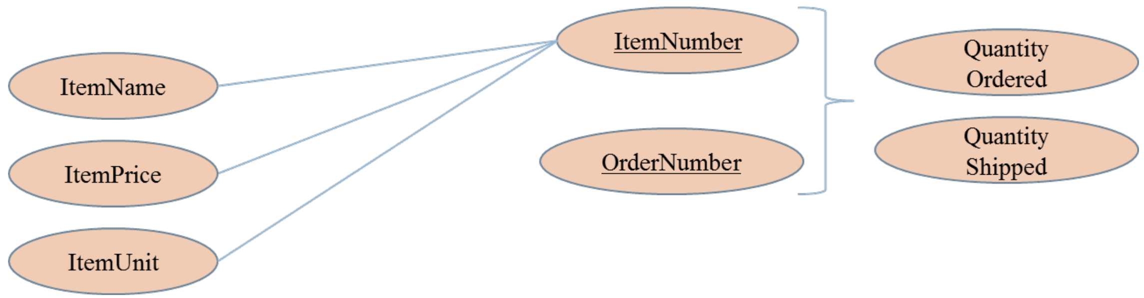
# Reason Anomalies Exist

- Several non-key attributes depend only on ItemNumber
- Should depend on the full primary key (ItemNumber + OrderNumber)

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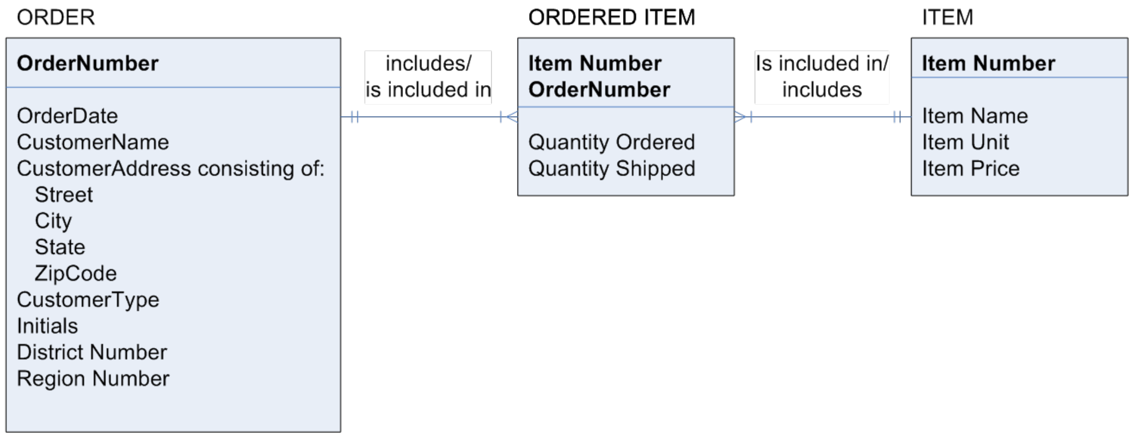
# Example: Second Normal Form

- For entities with concatenated keys…
- Do any attributes depend on just part of the key rather than the entire key?
- If yes, move partially-dependent attributes to a new entity…

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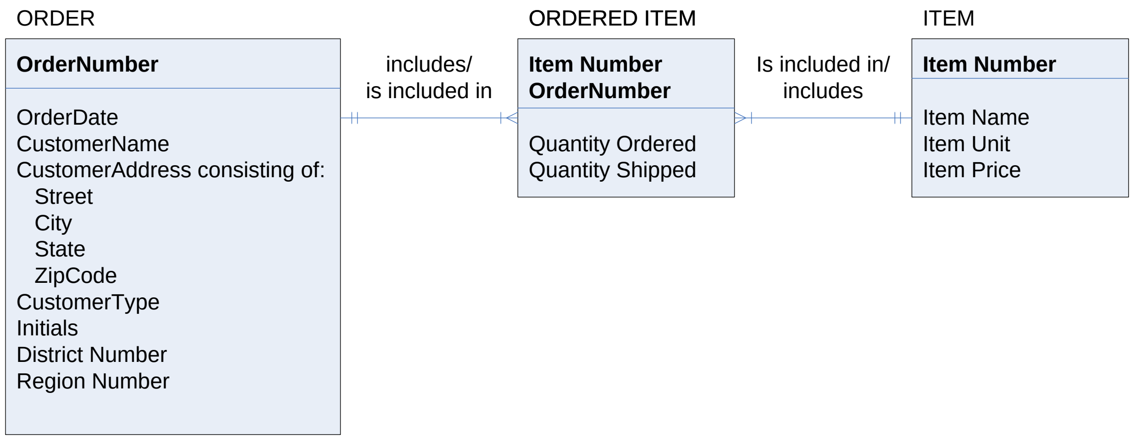
# Potential Anomalies with Second Normal Form

- The ORDER entity contains transitive dependencies
- Several non-key attributes depend on another non-key attribute, and NOT on the Primary Key
- CustomerAddress, CustomerType, DistrictNumber, and RegionNumber depend on the CustomerName, NOT the OrderNumber
- Solution: remove these attributes to a separate entity

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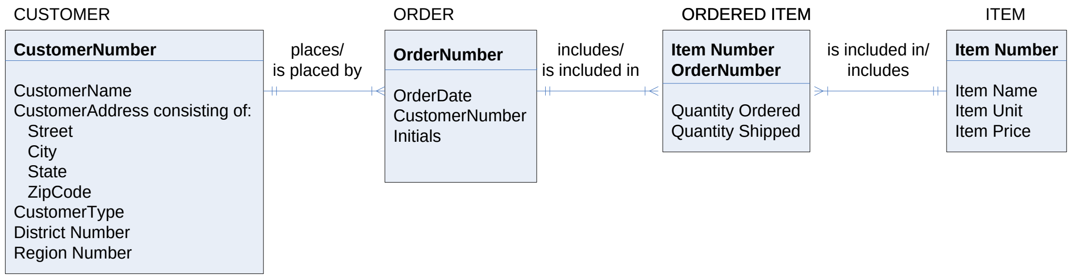
# Example: Third Normal Form

- Do any attribute values depend on an attribute that is not the entity's key?
- If yes, move these attributes to a new entity

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# Summary of Normalization Steps

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| | 1 Normal Form |
| Is the identifier composed of more than one attribute? If so, are any attribute values dependent on just part of the identifier? | Yes: Remove the partial dependency. Move the attributes to an entity in which their values are dependent on the entire identifier. Usually, you will need to create a new entity and add a relationship to connect the old and new entities.<br>No: The data model is in 2NF. |
| | 2 Normal Form |
| Do any attribute values depend on an attribute that is not the entity's identifier? | Yes: Remove the transitive dependency or derived attribute. Move the attributes to an entity in which their values are dependent on the identifier. Usually, you will need to create a new entity and add a relationship to connect the old and new entities.<br>No: The data model is in 3NF. |
| | 3 Normal Form |

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# Chapter Review

- Define the meaning and purpose of the entity and relationship shown on an entity relationship diagram (ERD).
- Explain the meaning and purpose of attributes included in a data model.
- Explain what is meant by an entity's identifier.
- Explain the meaning of the cardinality and modality of a relationship.
- Explain the concept of metadata and how it is compiled in the project repository.
- Discuss the process used to create a data model.
- Describe how to ensure that the process model and data model are balanced through the use of the CRUD matrix.
- Discuss how the normalization process is performed and how it contributes to the quality of the data model (from chapter appendix).

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# Key Terms

- 1:1 relationships
- 1:N relationships
- Assumptions
- Attribute
- Balance
- Business rules
- Cardinality
- Child entity
- Clients
- Concatenated identifier
- Create, read, update, delete (CRUD) matrix
- Data dictionary
- Data model
- Dependent
- Dependent entity
- Derived attributes
- Entity
- Entity relationship diagram (ERD)
- First normal form (1NF)
- Identifier
- Identifying relationships
- Independent entity
- Instances
- Intersection entity
- Logical data model
- Metadata
- M:N relationship
- Non-identifying relationship
- Normalization
- Parent entity
- Partial dependency
- Physical data model
- Relationships
- Repeating attributes
- Repeating attribute groups
- Second normal form
- (2NF)
- Subject area
- Third normal form (3NF)
- Transitive dependency

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